11 The discrete random variable X takes the values 0, 1, 2, 3, 4 and 5 with probabilities given by the formula

$$\mathrm{P}(X = x) = k(x + 1)(6 - x) .$$

(i) Find the value of k. **[2]**

In one half-term Ben attends school on 40 days. The probability distribution above is used to model X, the number of lessons per day in which Ben receives a gold star for excellent work.

(ii) Find the probability that Ben receives no gold stars on each of the first 3 days of the half-term and two gold stars on each of the next 2 days. **[2]**

(iii) Find the expected number of days in the half-term on which Ben receives no gold stars. **[2]**